

Apply the exponent laws to numerical expressions with integer exponents and extend to variables

Learning intention

- apply the exponent laws to numerical expressions with integer exponents and extend to variables.

Teach and model

- representing decimals in exponential form.
- relating simplification of expressions from first principles and counting to the use of exponent laws.
- relating the prefixes for SI units from pico- (trillionth) to tera- (trillion) to the corresponding powers of 10.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.